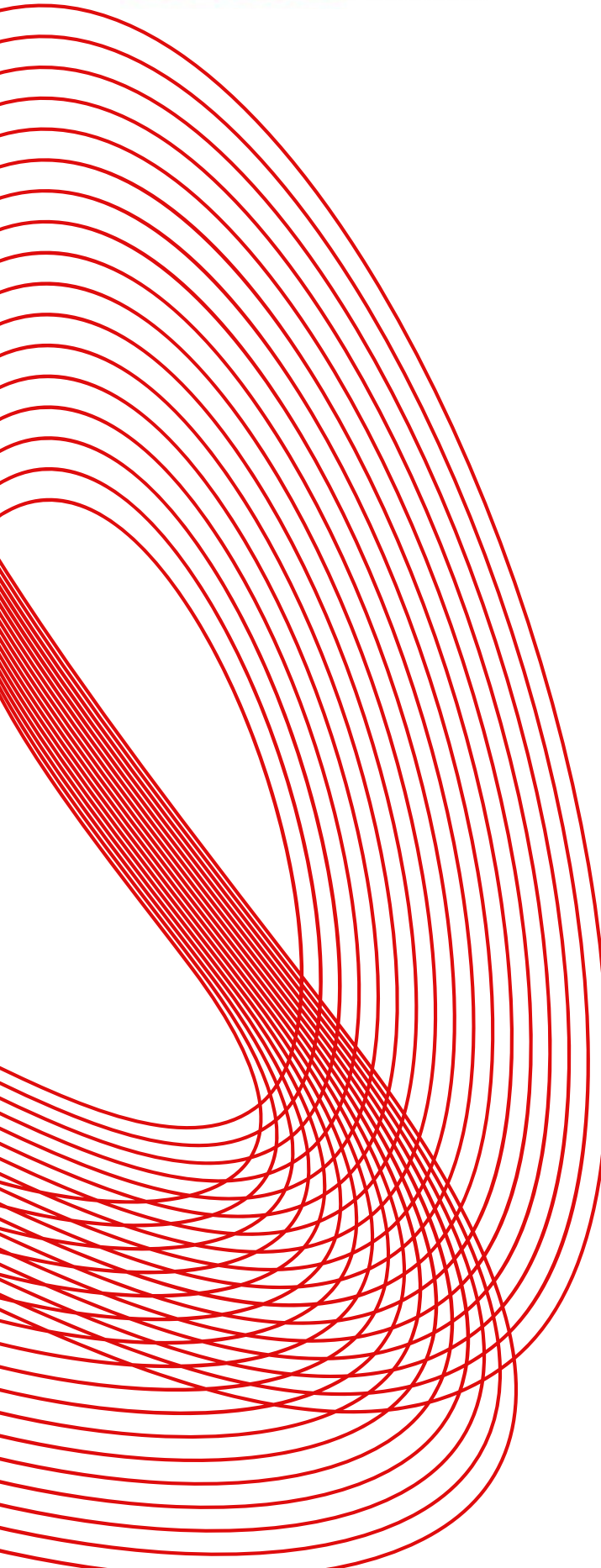


A large, abstract graphic on the left side of the page, composed of numerous concentric, overlapping red lines that form a dynamic, swirling shape, resembling a stylized 'S' or a series of nested ellipses.

S-IoT Project Report : Swarn Drone

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Promotion 2022-2027

Specifications

Objective

The purpose of this project is to fly a swarm of UAVs (2 to 3 UAVs) and try to lift off a payload using a net.

Functions

- UWB precise location system
- PID Controller for drone stabilization
- UWB Communication for high frequency refreshing data
- Make a simple back and forth keeping the drone formation (following a leader ?)

Deliverable

- Report with all progress, advancements and explanation of technical knowledge
- Functional UWB code
- Methodology and results of calibration of UWB sensors
- Bibliographical Studies
- Video of Drone demo
- Functional PID Corrector
- Functional ESP32 code implementation using Multi-threading

Bibliographical Studies

0.1 UWB

0.1.1 Definitions

Explain what is UWB

0.1.2 Algorithmes

Explain that we found and test different Algorithmes

Two way ranging (TWR)

Explain

Double Sided Two way ranging (DS-TWR)

Explain

0.2 Data treatment

Contenu de l'étude bibliographique...